

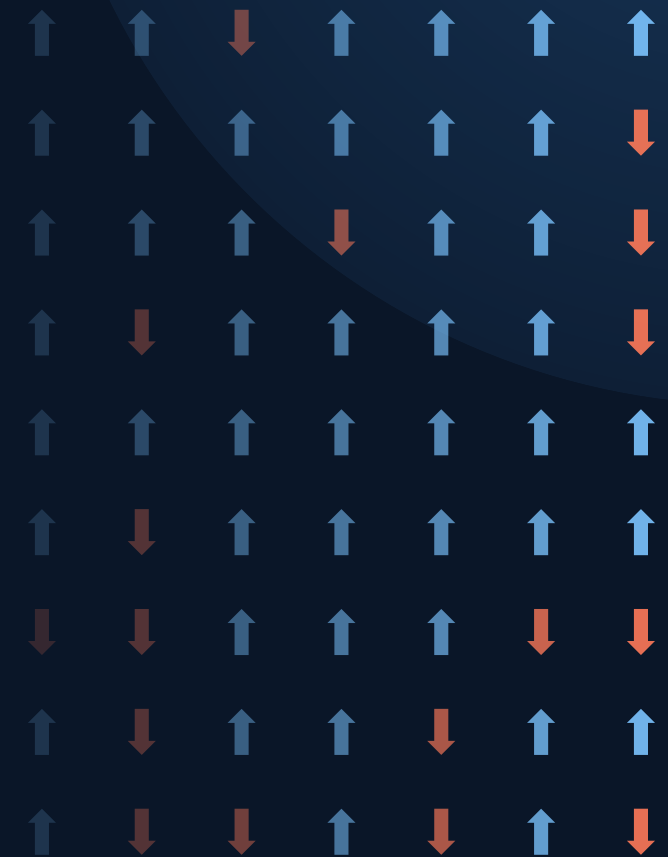
Measuring the External Field

Can we measure how hard a news article pushes on what people think?

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each arrow is one person's opinion

SCRIPT · slide 01 · about 1 min

Good morning, Dr. Songa, and thank you for the time. I'll take you through the dissertation as it stands after your review: what I set out to do, the tools and steps, what we found, where the proposal turned out wrong, what the work cannot claim, and the three papers. About fifteen minutes, and I'll end with what I need from you.

NUMBERS TO SAY

- About 15 minutes in total
- End with the four requests



Physics treats media as a push, h , on opinion. Nobody measures h

$$m = \tanh(\beta J m + h)$$

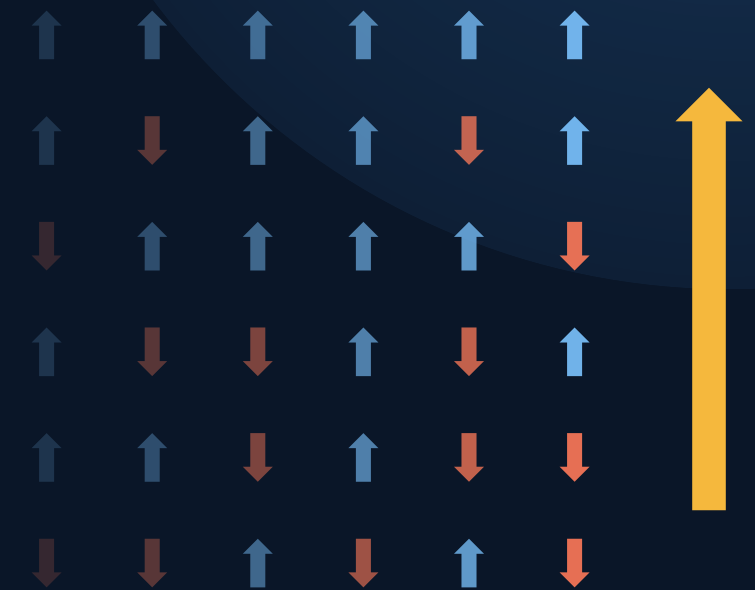
The mean-field opinion model: the same maths as magnets lining up.

m average opinion of the group: -1 everyone against, $+1$ everyone for.

J coupling: how strongly each person is pulled toward the people around them.

β how firmly people follow that pull instead of acting at random (one over the noise).

h external field: the push from media, the same on everyone.



Arrows are people. Gold arrow is media, pushing everyone the same way.

Every study we found picks h by hand. **Our question: can h be measured from the content itself?**

Physicists model a group forming an opinion the same way they model tiny magnets lining up. The equation is m equals $\tanh$ of $\beta J m$ plus h . m is the group's average opinion, from minus one, everyone against, to plus one, everyone for. J is how strongly each person is pulled toward the people around them, and β is how firmly they follow that pull instead of acting at random. h is the outside push from media, the same on everyone. In every study I read, h is chosen by hand. My question: can we measure h from the content itself?

NUMBERS TO SAY

- m : average opinion, -1 to $+1$
- J : pull toward neighbours
- β : how firmly people follow the pull
- h : the push from media



What the proposal assumed, and what testing the tools showed

THE PROPOSAL ASSUMED	WHAT WE FOUND	WHAT WE DID
We would measure the amygdala, the brain's alarm centre.	The model only predicts the brain's outer surface.	Used emotion-linked and reasoning-linked surface areas.
The score would be a ratio of the two areas.	Averages sit near zero, so the ratio broke for 69 of 400 articles.	Used a simple difference: emotion minus reasoning.
Outrage articles would stand out the most.	Outrage raised both areas by the same amount.	Reported it; fear is what stands out.
We would measure α (alpha), the strength of media's push.	The data could not pin it down.	Worked out the minimum α instead.
The score would detect manipulative articles.	Simple word counting did better.	Presented it as a measurement, not a detector.

We also corrected two coefficients in the proposal's physics equation. All changes are in the approved amendment.

Before the results, I want to be upfront about what changed from the proposal. Five assumptions turned out differently once we tested the tools. The biggest: we planned to measure the amygdala, the brain's alarm centre, but the model only predicts the brain's outer surface. So we used emotion-linked and reasoning-linked areas on the surface instead. All of this is in the amendment you approved.

- NUMBERS TO SAY**
- 5 changes, all in the approved amendment
 - Ratio broke for 69 of 400 articles
 - Two physics coefficients corrected



The tools we used, and why each one

TRIBE v2

Meta research, 2025

An AI model that predicts brain activity from text, sound and video. It learned from brain scans of adult volunteers watching TV and films.

Text-to-speech

reads each article aloud

The model was built on people listening, not reading, so every article is spoken first.

Word timing tool

forced alignment

Marks the moment each word is spoken, so the words and the sound line up for the model.

Brain area map

Glasser et al., 2016

A standard map that names areas of the brain's surface. We used it to pick emotion and reasoning areas.

400 public articles

four published collections

Anyone can download the same articles and check every number.

Cloud GPU + Python

free Kaggle Tesla P100

About 70 seconds per article. All code and results are public on GitHub.

No person was scanned. Every brain value in this work is a prediction from the model.

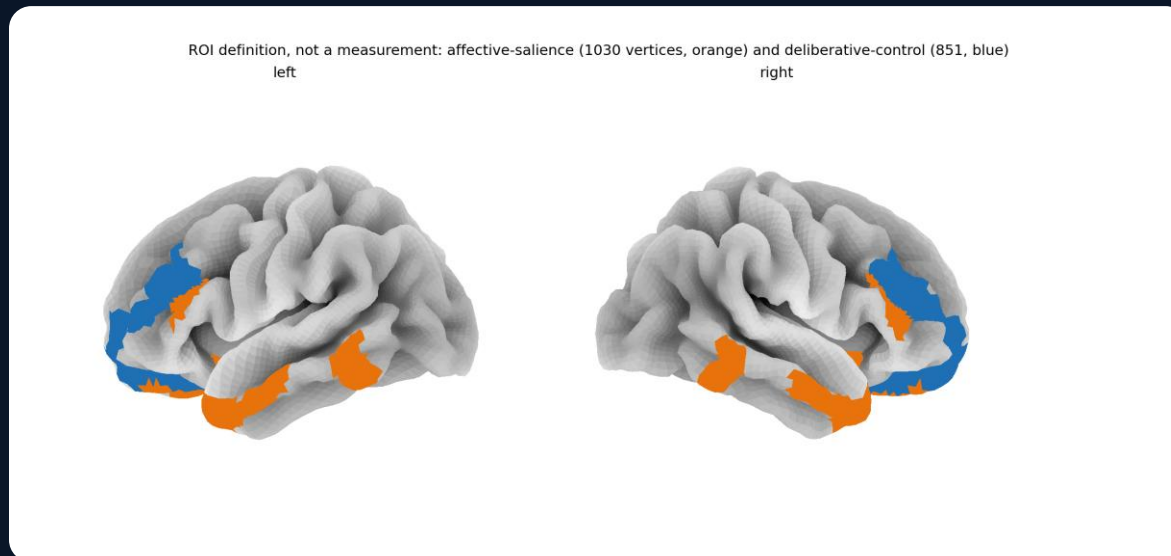
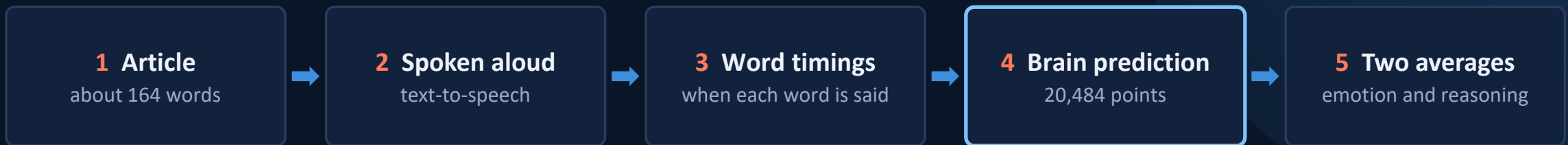
These are the tools. The main one is TRIBE v2, published by Meta's research lab. It learned from brain scans of adult volunteers watching TV and films, and it predicts how a typical brain would respond to new content. Because it was built on people listening, we read each article aloud first. Nobody was scanned in this project. Every brain value you'll see is a prediction.

NUMBERS TO SAY

- TRIBE v2: Meta, 2025
- About 70 seconds per article on a P100
- No person was scanned



How one article becomes one number



Orange: 1,030 emotion-linked points. Blue: 851 reasoning-linked points.

THE SCORE, X

$$X = \bar{A}_{\text{emotion}} - \bar{A}_{\text{reasoning}}$$

$\bar{A}$ = average predicted activity over that set of points.

Above 0: emotion areas predicted more active.

Below 0: reasoning areas predicted more active.

20,484 = points on a standard model of the brain's surface, where the prediction is made.

This is the whole process for one article. It's turned into speech, the timing of each word is marked, and the model predicts activity at 20,484 points across the brain's surface. We average 1,030 points in areas linked to emotion and 851 in areas linked to careful reasoning. The score is emotion minus reasoning. Above zero, emotion leads. Below zero, reasoning leads.

NUMBERS TO SAY

- 20,484 points on the brain surface
- 1,030 emotion, 851 reasoning
- Score = emotion minus reasoning



400 articles in four groups, matched in length

100

Fear-driven

fake news collection (ISOT)

100

Outrage

hyperpartisan news (SemEval-2019)

100

Clickbait

clickbait headlines (Webis-17)

100

Neutral

medical abstracts + real news

≈ 164

words per article, on average

What it means: All four groups are within 5 words of each other.

Why this value: So a longer article can't score differently just by being longer.

0.027

smallest effect the study could see

What it means: We sized the study before scanning a single article.

Why this value: 400 articles is enough to catch a difference as small as 2.7%.

× 2

every article scanned twice

What it means: Two separate runs on the GPU.

Why this value: Lets us measure how repeatable the score is.

We scored 400 articles from four public collections, 100 each: fear-driven fake news, outrage-style partisan news, clickbait, and neutral writing. We matched their lengths so no group wins by being longer, and we worked out before scanning that 400 was enough to see a small effect. Then we ran every article twice.

NUMBERS TO SAY

- 4 groups x 100 articles
- About 164 words each
- Smallest detectable effect 0.027
- Every article scanned twice



Result 1: the four groups really do differ

HOW MUCH THE GROUPS DIFFER (η^2)

0.107

$$\eta^2 = \frac{\text{variation between groups}}{\text{total variation}}$$

first run 0.107

second run 0.089

smallest we could see 0.027

The second run repeats the result. Both are far above what we could detect.

$\approx 11\%$

$\eta^2 = 0.107$: share of the differences explained by group

What it means: Knowing an article's group explains about 11% of why its score differs from the others.

Why this value: The groups overlap a lot; most variation is from one article to the next.

1 in a billion

$p = 1.0 \times 10^{-9}$: the chance this is luck

What it means: If the groups were truly the same, a pattern this strong would appear by chance about once in a billion tries.

Why this value: 400 articles and a consistent difference between groups.

First result: the four groups really do differ. The number is eta squared, 0.107. In plain terms, about 11 percent of the differences between articles line up with their group. The rest is article-to-article variation, because the groups overlap a lot. The chance of this being luck is about one in a billion, and the second run gave 0.089, so it held.

NUMBERS TO SAY

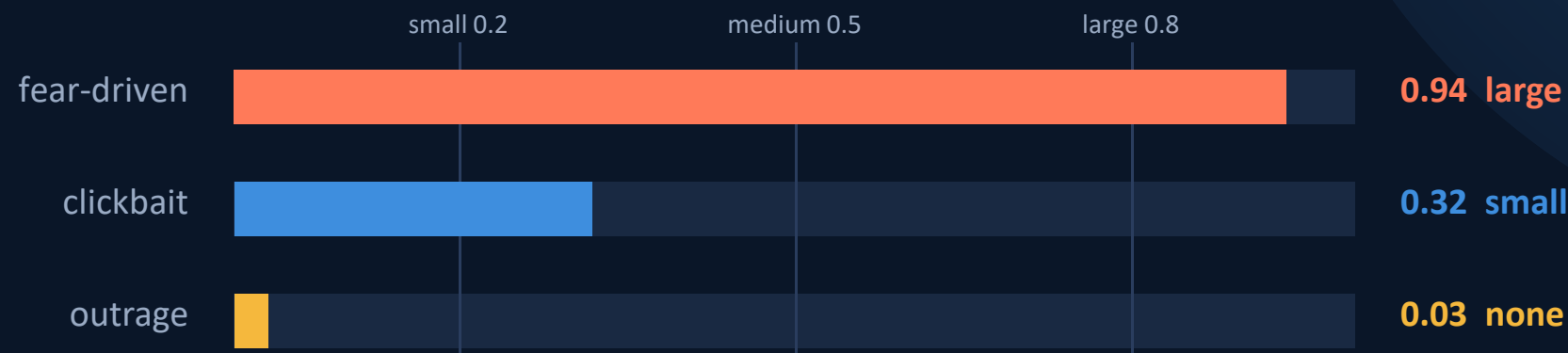
- 0.107 = about 11% explained by group
- p = about 1 in a billion
- Second run 0.089



THE RESULTS

Fear drives the difference; outrage barely moves the score

DIFFERENCE FROM NEUTRAL ARTICLES (COHEN'S d)



$$d = \frac{\bar{X}_{\text{group}} - \bar{X}_{\text{neutral}}}{s}$$

$\bar{X}$ = average score of a group. s = normal spread of scores.

What it means: 0.2 is small, 0.5 medium, 0.8 large. Fear articles sit almost one full spread away from neutral.

WHY OUTRAGE STAYS FLAT

Outrage raised emotion areas by 0.51 and reasoning areas by 0.49. The score is the gap between them, so it barely moves. Fear raised emotion areas only (0.61 vs -0.10).

THE RESULTS

08

SCRIPT · slide 08 · about 1 min

Which group drives it? Fear. Compared with neutral articles, fear-driven ones differ by 0.94, which counts as large. Clickbait is small. Outrage barely moves, which surprised us, because the proposal expected it to be strongest. Looking closer, outrage raises both the emotion and the reasoning areas by about the same amount, so the gap between them stays flat.

NUMBERS TO SAY

- Fear 0.94 (large)
- Clickbait 0.32 (small)
- Outrage 0.03: both areas rise, 0.51 and 0.49



THE RESULTS

The score repeats well for groups, not for single articles

0.87

Agreement between the two runs (ICC = 0.8725)

What it means: 1 means identical, 0 means no agreement. 0.87 is high.

Why this value: The speech voice and the computer's arithmetic vary a little between runs.

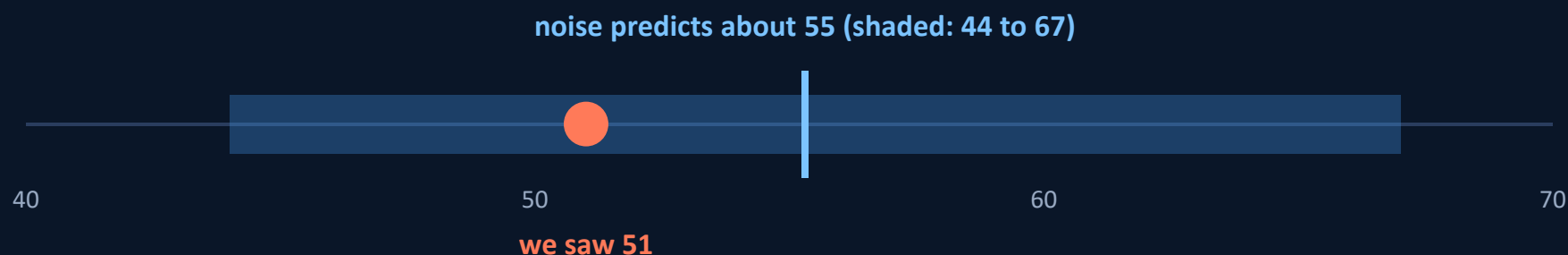
51 of 400

Articles whose score flipped sign (12.8%)

What it means: Noise alone predicts about 55. The ones that flipped all sit near zero.

Why this value: A score near zero tips either way with a tiny wobble, so we never judge one article.

FLIPPED ARTICLES: WHAT WE SAW AGAINST WHAT NOISE ALONE PREDICTS



Can we trust the score? We ran all 400 articles twice. Agreement between runs is 0.87, where 1 means identical, so it's high. It isn't perfect because the speech voice and the computer's arithmetic vary slightly. 51 articles flipped sign. Noise alone predicts about 55, and those articles all sit near zero. So we only make claims about groups, never one article.

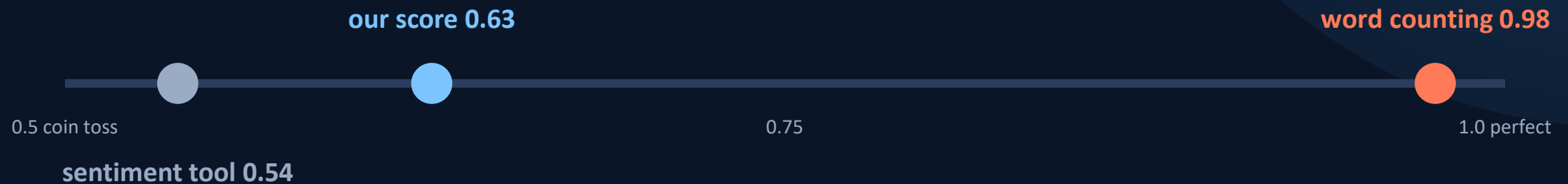
NUMBERS TO SAY

- Agreement 0.87 (1 = identical)
- 51 flipped, about 55 expected (44 to 67)



Honest check: simple word counting sorts articles better than our score

AUC = how well a score sorts manipulative from neutral articles. 0.5 is a coin toss, 1.0 is perfect.



WHY WORD COUNTING WINS

Each group came from a different source. From the words alone, a computer guesses the source 66% of the time, against 25% by chance. So it is partly spotting the source, not the style.

WHAT THIS MEANS FOR US

We cannot say the group differences come from writing style alone. Our score is offered as a measurement for the physics model, not as a manipulation detector.

Now the most important honest check. Simple word counting sorted manipulative from neutral articles far better than our score: 0.98 against 0.63, where 0.5 is a coin toss. The reason is that each group came from a different source, and the words give the source away. So we can't say the differences come from writing style alone. Our score is a measurement for the physics, not a manipulation detector.

NUMBERS TO SAY

- Word counting 0.98, ours 0.63, sentiment 0.54
- 0.5 = coin toss
- Source guessed 66% of the time vs 25% by chance



Does the brain model match real brains? Weakly, but yes

A company audit claimed this model is the opposite of real brains. We tested it on public brain scans of people watching the TV show Friends.

0.152

Best possible score (noise ceiling)

What it means: How closely real people's brains agree with each other watching the same episode.

Why this value: Brains differ and scanners are noisy, so even a perfect model can't score higher.

0.028

Model vs real brains (r), 2-minute clip

What it means: The model's predictions rise and fall with real brain activity, weakly.

Why this value: The best possible on that same clip is 0.096, so the model gets under a third of the way.

p = 0.048

Chance the match is luck

What it means: Just under the usual 5% cut-off.

Why this value: Two minutes of video is short. A longer run is needed to settle it.

Along the way we found and fixed three hidden bugs: the wrong episode loaded, a table read sideways, and two clocks out of step.

Does the brain model match real brains at all? A company audit claimed it was the opposite of real brain data, so we tested it on public scans of people watching the show Friends. Real people's brains agree with each other at 0.152, the best any model can reach. On a two-minute clip the model scored 0.028, against a best possible of 0.096 on that clip. Weak, but positive. We also found and fixed three hidden bugs. This is Paper 3.

NUMBERS TO SAY

- Best possible 0.152 (full episode)
- Model 0.028 vs 0.096 on the 2-minute clip
- p = 0.048
- 3 bugs found and fixed



The physics result: the minimum push media would need

$$h = \alpha X \Rightarrow \alpha \geq \frac{h_c(\beta J)}{\Delta X}$$

h the push from media on everyone (the field).

X our score for an article, emotion minus reasoning.

α strength: how hard one unit of score pushes. Not measured, so no value is quoted.

$h_c(\beta J)$ critical field: the smallest push that flips the majority. It grows with copying strength.

βJ copying strength. Above 1 a group settles on a majority by itself; 2 means strongly connected.

ΔX spread of our scores: from -0.070 to $+0.054$, so 0.124 .

$$\alpha \geq 4.29$$

Minimum strength when copying is strong ($\beta J = 2$)

What it means: Any study saying content like this flips opinion must use an α of at least 4.29, or the maths says it can't happen.

Why this value: Our scores cover a narrow range (0.124), so each unit of score must push hard.

WHAT IT DOES NOT SAY

It does not claim that media flips opinion. It is a bar any such claim has to clear. The more strongly people copy each other, the higher the bar.

Here is the physics result. We link our score to the model by h equals αX : the media push h is our score X times a strength, α . We couldn't measure α from the data, so we don't quote one. Instead we asked how big α must be to flip a majority. The model gives h_c , the critical field: the smallest push that flips the majority, which grows with the copying strength βJ . Our scores spread over ΔX , 0.124 . Dividing gives α at least 4.29 when βJ is 2 . Any future claim that content like this flips opinion has to clear that bar.

NUMBERS TO SAY

- $h = \alpha X$: push = strength $\times$ score
- h_c : smallest push that flips it
- $\Delta X = 0.124$ (-0.070 to $+0.054$)
- $\alpha \geq 4.29$ at $\beta J = 2$
- No value of α is claimed



Can it analyse any content, even what children watch?



WHAT IT CAN DO

- Take video and sound as well as text, so a cartoon or an advert can be scored.
- Compare groups of content, like two channels or two kinds of programme.
- Score the content, never the viewer. Nobody is scanned.



WHAT IT CANNOT DO YET

- Say how a child reacts. The model learned from adult brains, and children's brains are still developing.
- Speak for video: only news text read aloud was tested here.
- Rate one show: about 1 in 8 items flip between runs.



WHAT IT WOULD TAKE

- Brain scans from children, collected with ethics approval and consent.
- A test set of children's videos.
- Content where source and type are mixed.
- Safeguards so it is never used to target children.

Short answer: technically possible, but not yet valid for children.

At my first presentation I was asked whether this could analyse any content, even what children watch. Technically, yes: the pipeline takes video and audio, so it could score a cartoon or an advert. But the model learned from adult brains, and children's brains are still developing, so it can't say how a child reacts. We also only tested news text. For children, we'd need child brain data and ethics approval first.

NUMBERS TO SAY

- Can: video and audio, groups of content
- Cannot: adult-trained model, 1 in 8 items flip
- Needs: child data and ethics approval



What this work cannot claim yet

LIMIT	WHY IT MATTERS	NEXT STEP
Predictions, not brain scans	Nobody was scanned; the model could be wrong.	The real-brain check (Paper 3).
Group is mixed up with source	Each group came from one collection.	A new set where every source has every type.
Brain surface only	No amygdala or other deep areas.	A model that predicts deep areas.
Strength of the push (α) unknown	The data can't pin it down.	Use the minimum rule instead.
The real-brain check is short	Only 2 minutes of video so far.	Longer clips, run one at a time.

To be clear about the limits: every brain value is a prediction, not a scan. Group and source are mixed together in this dataset. The model can't see deep areas like the amygdala. We couldn't measure alpha. And the check against real brains used only two minutes of video. Each is stated in the thesis, and each points to the next piece of work.

NUMBERS TO SAY

- 5 limits, each stated in the thesis



Three papers, and what I need from you today

PAPER 1 · Physica A

The minimum push

How strong media would have to be to flip a majority. Pure physics; needs no brain data.

12 pages, ready

PAPER 2 · Physica A

The measuring tool

The process, the 400 articles, the results, and the word-counting comparison.

16 pages, ready

PAPER 3 · Imaging Neuroscience

Does the model match real brains?

The best-possible score, the three bugs, and the first check.

8 pages, one longer run to go

TODAY I AM ASKING FOR

- 1 Any corrections you need before the thesis is final.
- 2 Your signature on the declaration page (page iii).
- 3 Clearance to deposit with the library (bound copy by Thursday 24 September).
- 4 Your view on co-authorship of the three papers.

Three papers come out of this. Paper 1 is pure physics: the minimum-push rule, no brain data needed. Paper 2 is the measuring tool and the 400 articles, including the word-counting comparison. Paper 3 is the check against real brains. The dissertation comes first. I'd like any corrections you need, your signature on the declaration page, and clearance for the library by Thursday. And I'd like to ask about co-authorship on the papers.

NUMBERS TO SAY

- Bound copy to the library by Thu 24 Sep
- Graduation clearance by Mon 28 Sep
- Ask about co-authorship

Many of these grow out of the first presentation's question about children's content

Q1 Can this analyse any content, even what children watch?

A Technically it can score video and audio, not only text. But the model learned from adult brains, so it can't tell us how a child reacts, and we only tested news text. It would need child brain data and ethics approval first.

Q2 Could a parent or a regulator use it to rate one show?

A Not yet. About 1 in 8 items flip between runs, so a single score isn't reliable. It's only trustworthy for comparing groups of content, and not for children until it has been tested on them.

Q3 Was anyone scanned? Is it reading people's minds?

A No. Nobody was scanned. The scores are predictions from a published model of a typical adult brain. The tool rates content, not people, and knows nothing about any individual.

Many of these grow out of the first presentation's question about children's content

Q4 Could someone use it to make content more manipulative?

A That's a real risk with any measurement of content. That's why it's framed as a research measurement, reports only group results, and the model's licence allows non-commercial research only.

Q5 If word counting does better, why not use that?

A Word counting answers a different question: it mostly learns which source an article came from. Our score is meant to be the push in the opinion model. The comparison is there to be honest about what the score is not.

Q6 Why read the articles aloud?

A The model was trained on people watching and listening, so it expects speech with word timings. Reading aloud puts the article in the form the model understands.

Many of these grow out of the first presentation's question about children's content

Q7 Why no amygdala, when the proposal named it?

A The released model only predicts the brain's outer surface. The amygdala sits deep inside, so we can't measure it and make no claim about it. We used surface areas linked to emotion instead.

Q8 What would it take to work for children's media?

A Brain scans from children collected with ethics approval and consent, a test set of children's videos, content where source and type are mixed, and a longer check against real brains.

Q9 Why is this a physics project?

A The opinion model is statistical physics, the same maths as magnets lining up. The contribution is measuring the push in that model and working out the minimum strength it needs. The brain model is the measuring tool.

Many of these grow out of the first presentation's question about children's content

Q10 What does 'alpha at least 4.29' mean in practice?

A It's a bar to clear. If someone claims content like this flips a strongly connected group's opinion, their alpha must be at least 4.29, or the model says it can't happen. It does not say media does flip opinion.

Q11 Why only 400 articles?

A We calculated before scanning that 400 was enough to see an effect as small as 2.7%. Each article takes about a minute of GPU time, and we ran all of them twice.

Q12 Is it okay to use Meta's model?

A Yes, for research. It's released under a non-commercial licence, it's cited throughout, and this project makes no money from it.

If a symbol comes up, say it like this

m Average opinion of the group, from -1 (all against) to $+1$ (all for).

J, coupling How strongly each person is pulled toward those around them.

β , beta How firmly people follow that pull instead of acting at random.

h, field The push from media, the same on everyone. $h = \alpha X$.

X, our score Emotion-area activity minus reasoning-area activity, predicted.

η^2 , eta squared Share of the differences explained by group. 0.107 is about 11%.

p-value Chance a result this strong appears by luck. 1×10^{-9} is about one in a billion.

d, Cohen's d Size of a difference in units of normal spread. 0.2 small, 0.5 medium, 0.8 large.

ICC Agreement between two runs. 1 identical, 0 none. Ours 0.87.

AUC How well a score sorts two kinds of article. 0.5 coin toss, 1 perfect.

r, correlation How closely two things rise and fall together. 0 none, 1 perfect.

Noise ceiling Best score any model can reach, given how much real brains differ. Ours 0.152.

α , alpha Strength of media's push per unit of our score. Not measured.

β_J How strongly people copy each other. Above 1, a group settles on a majority by itself.

ΔX Spread of our scores, lowest to highest: 0.124.

h with small c The critical push: what it takes to flip a majority.